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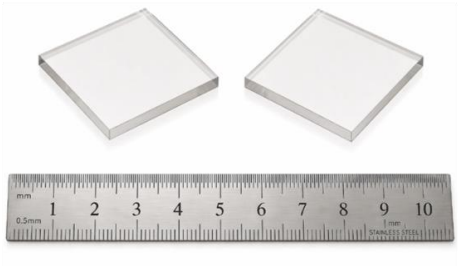
+91 8928373100 info.cctechmumbai@gmail.com Navi Mumbai, India

Our Product Range

At Carbon Carat Technology LLP, we specialize in advanced wide bandgap semiconductor materials, combining exceptional purity, cutting-edge engineering, and sustainable innovation. From electronic-grade diamonds (boron- and phosphorus-doped) to quantum-grade diamond wafers, our solutions are engineered to exceed the highest industry standards.

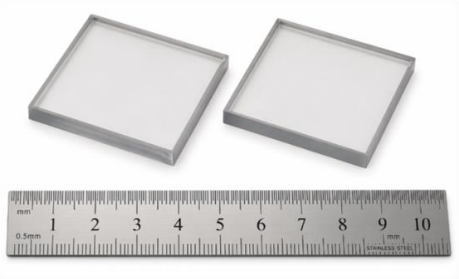
Boron doped diamond plates

(p-type, high purity, precision doping)

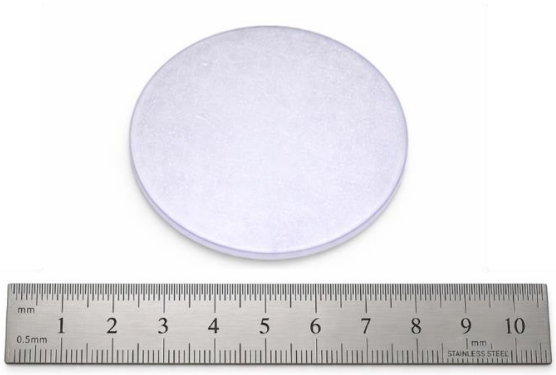


Phosphorous doped diamond plates

(n-type, high purity, controlled doping)

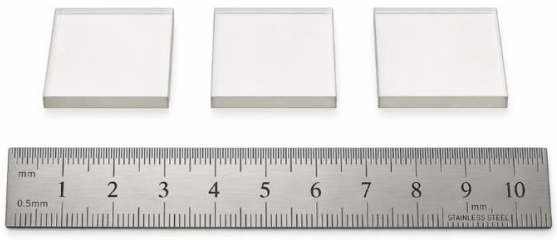


Optical grade polycrystalline diamond wafer



Semiconductor grade diamond plates

(Ultra-high purity, atomically smooth surface)



6 kW MPCVD Diamond Deposition System

Applications

- High-quality electronic, optical and quantum grade single crystal diamond growth
- Polycrystalline & free-standing diamond films
- Advanced carbon nanomaterials: graphene, CNTs, fullerenes

Key Features

- High power-density 6 kW, 2.45 GHz microwave plasma
- Water-cooled stage & chamber for stable long-duration operation
- Plasma self-heating for substrate temperature control
- Full-range vacuum gauge for precise pressure monitoring
- Turbo + rotary pumps (up to 1×10^{-5} Pa) with automatic pressure control
- Efficient cooling system for high-power reliability
- 24” monitor with PLC automation (stores more than 20 recipes)
- Fully automated process module for consistent, high-quality growth

Warranty

- 12 months from installation or 14 months from shipment (whichever earlier)
- Excludes misuse, unauthorized modifications, and consumables
- Site utilities readiness required; delays do not extend warranty

➢ **Reliable, automated, and precision-driven system for advanced diamond and carbon material synthesis.**



10kW/12kW MPCVD Diamond Deposition System

Applications

- High-quality electronic, optical and quantum grade single crystal diamond growth
- Polycrystalline & free-standing diamond films

Key Features

- 10kW/12kW high power-density microwave plasma
- 2.45 GHz magnetron for efficient plasma generation
- Wide plasma range: low to ultra-high power density with temperature control
- Recipe-driven computer control with pyrometer (400–1700 °C)
- Water-cooled aluminium structure for stability
- 5 MFC gas lines (expandable on request)

Warranty

- 12 months from installation or 14 months from shipment (whichever earlier)
- Excludes misuse, unauthorized modifications, and consumables
- Utility delays (gas, power, water) do not extend warranty

➢ **High-power, precision-controlled system for scalable diamond and advanced carbon material synthesis.**



915 MHz MPCVD System (30 kW)

A next-generation, high-power MPCVD platform engineered for large-area, high-quality diamond and advanced carbon material growth with exceptional process stability and control.

Applications

- High-quality single crystal diamond growth
- Polycrystalline & free-standing diamond films
- Advanced carbon nanomaterials: graphene, CNTs, fullerenes

Key Features

- 30 kW high power-density microwave plasma
- 915 MHz magnetron for deep plasma penetration & uniform growth
- Wide operating range: low to ultra-high plasma density with precise temperature control
- Recipe-driven automation with pyrometer (400–1700 °C)
- Water-cooled aluminium structure for continuous, stable operation
- Multi-gas MFC system (5 lines, expandable on request)

Warranty

- 12 months from installation or 14 months from shipment (whichever earlier)
- Excludes misuse, unauthorized modifications & consumables
- Utility readiness required; delays do not extend warranty

➢ **Built for high-throughput, large-scale, and precision-driven diamond synthesis.**

Hot Filament CVD (HFCVD) Diamond System

A proven and cost-effective technology for high-quality diamond and advanced carbon coatings, combining simplicity, reliability, and industrial scalability.

Applications

- Nanodiamond and microcrystalline diamond coatings
- CVD diamond films
- Graphene, CNTs, and advanced thin films

Key Features

- Mature, industry-trusted technology with simple and robust design
- High-performance deposition with excellent safety and reliability
- Double-layer stainless steel chamber for durability and thermal stability
- Compact, two-stage structure for efficient operation
- Integrated gas & vacuum system with optimized pipeline design
- Real-time monitoring and precise control of gas flow and pressure
- Fully automated control system for consistent, repeatable results

➢ **Reliable, scalable, and efficient solution for industrial diamond and carbon coating applications.**

Thermal CVD System

Thermal Chemical Vapor Deposition (CVD) is a leading method for synthesizing high-quality, large-area graphene by decomposing hydrocarbon precursors (e.g., methane) at high temperatures (~650 - 1000°C) on catalyst substrates like Cu. It is ideal for depositing high-quality conductive, semiconductor, ferroelectric, and optical films, with excellent capability for evaporating refractory metals.

Applications

- Graphene, single-walled (SWCNTs) and multi-walled (MWCNTs) growth.
- Thin-film coatings for advanced materials
- Micro–nano device fabrication and processing.
- Research and small-scale production

Substrate Compatibility

- Rigid: glass wafers, silicon wafers, diamond wafers, etc.
- Flexible: PDMS, PTFE, PI

➢ **Enabling uniform, and high quality graphene and CNT growth and precise coating over both rigid and flexible substrates.**

Magnetron Sputtering Coating System

A versatile thin-film deposition platform designed for precision coating of metals, semiconductors, and dielectric materials. Widely used in semiconductor, LED, photovoltaic, and advanced materials research, the system supports both R&D and small-batch production.

Applications

- Metal, semiconductor, and dielectric thin-film deposition
- Semiconductor devices, LEDs, and photovoltaic coatings
- Research laboratories and pilot-scale manufacturing

System Highlights

- Single-chamber, compact design for stable and efficient operation
- Integrated magnetron sputtering source with ion bombardment capability
- Rotating substrate holder with optical heating for uniform film growth
- Complete vacuum, gas flow, water-cooling, and safety alarm systems
- High-reliability sputtering power supply for consistent deposition

Automation & Control

- PLC-based intelligent control system
- Touchscreen interface for full process management
- Automated vacuum, sputtering, gas filling, and venting operations

➢ **Engineered for high-uniformity coatings, process repeatability, and user-friendly automated operation.**

Electron Beam Evaporation Coating System

A high-precision thin film deposition system designed for advanced material engineering. It integrates a vacuum evaporation chamber, E-type electron gun, thermal evaporation source, rotating heated substrate holder, gas flow control, high-efficiency vacuum system, and intelligent electronic controls—all built on a robust platform for stable and reliable operation.

Applications
Engineered for the deposition of high-quality conductive, semiconductor, ferroelectric, and optical films. Ideal for research laboratories, academic institutions, and small-scale production environments.

- **Delivering precise, uniform, and high-performance thin film coatings for cutting-edge applications.**



Technical Articles

Microwave Plasma CVD System from SCD Growth

Simulation, development, and optical emission spectroscopy of microwave plasma chemical vapor deposition reactor for synthesis of large-area single crystal diamond, *Next Materials* 8 **(2025)** 100791.

Dependency of Deposition Parameters on Morphology of the Single Crystal Diamond Films

Step growth in single crystal diamond grown by microwave plasma chemical vapor deposition, *Diamond & Relat. Mater.* 15 **(2006)** 304.

CARBON CARAT TECHNOLOGY LLP (CCT)

Founded in 2021 by a team of professional engineers, CCT is a high-tech enterprise specializing in advanced thermal and CVD equipment. We design and manufacture standard and customized furnaces and related systems, delivering high quality at competitive cost.

What We Offer

- High-temperature CVD equipment (700–1200 ℃, extendable up to 1700 ℃)
- Customized solutions tailored to customer requirements
- Strong manufacturing base with reliable quality control

Our Strengths

- Expert team & rich industry experience
- Safe, high-quality materials
- Fast response and customer-focused approach

➢ **Driven by professionalism and innovation, we deliver reliable equipment and solutions to clients worldwide.**



Team Work



High-quality



Safe Materials



Rich Experience

Contact Us

We'd Love to Hear from You

Have a question? Reach out via email or connect with us directly.



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- **We're here to assist you with quick and reliable support.**